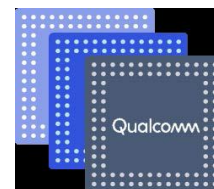


NR5GSub6_single_chain_2rx_4rx_test_application_note

KBA- 200105221851
Rev.1



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Revision History

Revision	Date	Description
1	2020-1-6	Initial release
2		
3		

Qualcomm
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2020-07-31 00:44:04 PDT
linyz4@motorola.com

NR5GSub6_single_chain_2rx_4rx_test_application

- KBA number: KBA-200105221851
- Applicable platform: SM7250/SM8250
- Issue/problem description:
 - Guide the customer to do single chain/2rx/4rx sensitivity test.
- 问题现象:
 - 指导客户测试单rx/2rx/4rx sensitivity

NR5GSub6_single_chain_2rx_4rx_test_application

- **1. Proper HW connection by divider for sub6:**
 - a. PRX: connecting PRX ant while testing PRX
 - b. DRX: connecting PRX+DRX ants while testing DRX
 - c. PRXMIMO: connecting PRX+PRXMIMO ants while testing PRXMIMO
 - d. DRXMIMO: connecting PRX+DRXMIMO ants while testing DRXMIMO
 - e. 2RX: connecting PRX+DRX ants while testing 2RX
 - f. 4RX:connecting PRX+DRX+PRXMIMO+DRXMIMO ants while testing 4RX
- **2. EFS file settings for controlling a specific Rx chain, place the necessary EFS file into the User Equipment (UE) EFS folders, under the following directory:**
 - a. for single chain test, EFS file with filename rfmgr_set_single_chain_rx should be put into below folder: /nv/item_files/modem/nr5g/ML1/
 - b. for 2Rx/4Rx chain test, EFS file with filename num_rx_cfg should be put into this folder: /nv/item_files/modem/nr5g/RRC/

Note:

- 1) EFS file location(refer to attached file named with “RX_sensitivity_EFS”)

PRX+DRX as 2Rx chain: RX_sensitivity_EFS\2Rx\RRC_NUMRX_2

PRX+DRX+MIMO+MIMO as 4Rx chain: RX_sensitivity_EFS\4Rx\RRC_NUMRX_4

RX0 as PRX single chain: RX_sensitivity_EFS\single chain\chain_rx0

RX1 as DRX single chain: RX_sensitivity_EFS\single chain\chain_rx1

RX2 as PRXMIMO single chain: RX_sensitivity_EFS\single chain\chain_rx2

RX3 as DRXMIMO single chain: RX_sensitivity_EFS\single chain\chain_rx3

- 2) We can create specific file folder such nr5g/ML1/RRC if no this folder in default EFS file on UE

NR5GSub6_single_chain_2rx_4rx_test_application

- 3. Proper test steps

Steps	Actions
1	Load EFS into devices and reboot
2	Start online call
3	Specific Rx chain triggered automatically based on EFS file configuration
4	Run this specific chain RF test
5	Hand over to other channel/band
6	Rx chain state remains the same
7	Run Rx chain RF test
8	End call

Note:

Callbox setting(refer to attached file named with callbox setting for Rx sensitivity test):

Coreset symbol duration: 2

Num Coreset RB: 48

AL8 : 1

CCE index allocation type : Auto

Questions?

For additional information or to submit technical questions, go to:

<https://createpoint.qti.qualcomm.com>

File a case with the proper problem area code to ensure that the problem assignment is sent to the correct support team for quicker resolution.

Problem Area Code	Support Team
Case Record Type	Wireless Device Support
Problem Area 1	RF Hardware (cellular)
Problem Area 2	Transceiver or RFFE
Problem Area 3	-